

Regime-Aware Portfolio Optimization via Offline Reinforcement Learning with CVaR Reward: Evidence from the Brazilian Stock Market

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Abstract

The Brazilian stock market is subject to pronounced macroeconomic cycles driven by monetary policy, making it an environment where regime-aware portfolio strategies hold particular promise. This paper asks whether an offline reinforcement learning agent trained with a Conditional Value-at-Risk (CVaR) reward and macroeconomic state variables produces portfolios with superior risk-adjusted performance out-of-sample relative to classical benchmarks on the Brazilian stock exchange (B3). We formulate portfolio rebalancing as a Markov Decision Process, train a Conservative Q-Learning (CQL) agent on an offline dataset spanning 2016–2021 whose state encodes SELIC regime indicators alongside asset-level return dynamics, and evaluate it on a held-out window from 2022 to 2024 against Equal-Weight, Minimum-Variance and Risk-Parity portfolios measured on the same 720 trading days.

The answer is negative, and the ablation localizes why. The full proposed configuration attains a Sharpe ratio of 0.4058 and $\text{CVaR}_{0.95}$ of 0.0260 against 0.6076 and 0.0226 for the equal-weight portfolio, which dominates it on every metric reported. Decomposing the model one component at a time, the CQL conservatism penalty contributes +0.29 in Sharpe and is the only component that helps; the CVaR reward costs -0.24 and, contrary to its purpose, worsens realised tail loss at every step where it is active; the macro state recovers +0.20 but leaves the full model below the simpler Sharpe-reward configuration that omits both. We trace the reward failure to its construction rather than to the market: the CVaR signal scores candidate weights against the same twenty-day window the state already encodes, making it a deterministic function of the observation and therefore no feedback at all. Regime-conditional results invert the motivating hypothesis, with the agent’s relative performance worst in tightening phases and positive only while rates fall. Measured turnover of 38 to 88 times portfolio value per year, against roughly one for

the benchmarks, places plausible transaction costs at the same order of magnitude as the entire return under comparison. We report the $1/N$ puzzle as holding in this setting and identify the reward specification, not the offline paradigm, as the component requiring redesign.

Keywords: offline reinforcement learning; portfolio optimization; CVaR; conservative Q-learning; Brazilian stock market; regime detection.

1 Introduction

The Brazilian equity market is among the most volatile in the emerging-market universe, where returns are shaped not only by firm-level fundamentals but also by sharp swings in the domestic monetary policy cycle. Between 2021 and 2023, the Banco Central do Brasil raised the benchmark SELIC rate by 523 basis points—from 2.00% to 13.75%—the most aggressive tightening cycle in two decades, triggered by a combination of post-pandemic supply shocks and persistent inflationary pressure. This episode illustrates a structural feature of the Brazilian market: asset-return dynamics shift substantially across interest-rate regimes, with aggregate Sharpe ratios varying from -0.23 during rising-rate periods to $+0.88$ during stable-rate periods in our sample. Classical portfolio optimization frameworks, however, are largely regime-agnostic. Mean-variance optimization (Markowitz, 1952) and its variants estimate a single covariance structure over the entire sample, ignoring the non-stationarity induced by monetary-policy transitions. Rule-based approaches such as the $1/N$ strategy (DeMiguel et al., 2007) and Risk Parity (Qian, 2011) are robust to estimation error but equally insensitive to prevailing macroeconomic conditions.

A natural extension is to use reinforcement learning (RL), which can, in principle, learn a state-dependent policy that adapts asset weights to the current economic environment. Online RL agents trained on rolling windows of live market data have been applied to portfolio management with promising results (Jiang et al., 2017), but this paradigm introduces a subtle methodological problem: the agent can implicitly learn from future information during training because the reward is computed on the same data stream that updates the policy, creating a look-ahead bias that inflates reported performance. Offline RL (Levine et al., 2020) resolves this by training entirely on a fixed, historical dataset—the offline dataset—and deploying the learned policy only on held-out data. Conservative Q-Learning (CQL) (Kumar et al., 2020) is a state-of-the-art offline RL algorithm that avoids the overestimation of Q-values for out-of-distribution actions by adding a conservatism penalty to the Bellman loss, making it particularly well suited to the noisy, non-stationary nature of financial time series.

Despite the growing body of work applying RL to asset allocation, three important gaps remain in the literature. First, most studies focus on developed markets; evidence from the Brazilian market, with its distinctive macro-financial dynamics, is scarce. Sec-

ond, the predominant reward signal is the Sharpe ratio, which treats upside and downside volatility symmetrically and can incentivize the agent to accept large tail losses. Third, offline RL methods have rarely been combined with explicit macroeconomic regime information in the state representation, leaving the potential of regime-conditional portfolio decisions largely unexplored.

This paper addresses all three gaps. Our specific contributions are as follows:

1. **Regime-aware state representation.** We construct a 225-dimensional state vector that combines asset-level return statistics with three macro-financial inputs: the SELIC target rate, the average bank lending spread to households, and a one-hot encoding of the SELIC regime as rising, stable, or falling. The encoding lets the agent condition portfolio decisions on the prevailing monetary-policy stance rather than on price history alone.
2. **CVaR reward function.** We replace the standard Sharpe-based reward with a composite signal that penalizes tail risk via the Conditional Value-at-Risk at the 95% confidence level while retaining a linear expected-return component weighted by a risk-aversion parameter $\lambda = 0.5$. This incentivizes the agent to control downside risk explicitly, a property especially valuable in the fat-tailed return distributions observed in emerging markets.
3. **Rigorous out-of-sample evaluation.** The classical benchmarks are evaluated under a rolling walk-forward protocol with 35 quarterly folds, re-estimating weights on 252-day windows and measuring on the following 63 days; the offline agent is trained once on 2016–2021 and deployed frozen over a 2022–2024 holdout. Neither protocol admits look-ahead bias, and Section 4 is explicit about the asymmetry between them. We benchmark the CQL agent against Equal-Weight, Minimum-Variance, and Risk-Parity portfolios and conduct an ablation study that isolates the contributions of the offline paradigm, the CVaR reward, and the macro state variables.

The remainder of the paper is organized as follows. Section 2 reviews the related literature on classical portfolio optimization, reinforcement learning in finance, offline RL, CVaR-based objectives, and macroeconomic regime detection. Section 3 describes the dataset, data sources, and regime classification. Section 4 presents the MDP formulation, the CQL algorithm, the reward function, and the evaluation protocol. Section 5 reports out-of-sample results and the ablation study. Section 6 interprets the findings and discusses limitations. Section 7 concludes.

2 Literature Review

2.1 Classical Portfolio Optimization

The foundation of quantitative portfolio management is the mean-variance framework introduced by Markowitz (1952), which characterizes optimal portfolios as those lying on the efficient frontier—the set of allocations that maximize expected return for a given level of variance, or equivalently minimize variance for a target return. Despite its theoretical elegance, mean-variance optimization is notoriously sensitive to estimation error in the sample mean and covariance matrix, which causes the optimal weights to be unstable and the out-of-sample performance to deteriorate significantly (Merton, 1980).

DeMiguel et al. (2007) conduct an extensive empirical study of fourteen mean-variance extensions across seven datasets and find that none consistently outperforms the naive $1/N$ equal-weight portfolio in terms of Sharpe ratio or turnover out-of-sample. This result—often called the $1/N$ puzzle—suggests that estimation error in classical methods is severe enough to erode all theoretical benefits from optimization. The Minimum-Variance Portfolio (MVP), which ignores the mean and relies solely on the covariance structure, partially mitigates this problem and tends to outperform full mean-variance models, though it still underperforms simple equal-weight strategies in many settings (Clarke et al., 2011).

Risk Parity is a more recent alternative that allocates weights inversely proportional to individual asset volatility contributions, so that each asset contributes equally to total portfolio risk (Qian, 2011; Maillard et al., 2010). Risk Parity gained practical traction after the 2008 financial crisis because its diversification principle is independent of return estimates. However, like MVP and equal-weight, Risk Parity is entirely regime-agnostic: it applies the same weighting logic regardless of whether the economy is tightening, easing, or at a neutral stance. In emerging markets with pronounced monetary cycles, this insensitivity to the macro environment is a material limitation.

2.2 Reinforcement Learning in Finance

The application of reinforcement learning to financial decision-making has grown rapidly since the introduction of deep Q-networks (Mnih et al., 2015), which demonstrated that neural network function approximators could learn superhuman policies directly from raw inputs. Early applications to portfolio management used Q-learning on discretized action spaces (Moody and Saffell, 1998), but the continuous nature of portfolio weights motivated a shift to policy-gradient and actor-critic methods.

Jiang et al. (2017) propose a Convolutional Neural Network-based portfolio management system trained with a policy-gradient algorithm, demonstrating consistent outperformance over buy-and-hold and equal-weight strategies on cryptocurrency markets. Ye et al. (2020) apply the Proximal Policy Optimization (PPO) algorithm to U.S. equity

portfolios and show that including technical indicators in the state space improves risk-adjusted returns. Deep Deterministic Policy Gradient (DDPG) (Lillicrap et al., 2015) has been widely adopted for its ability to handle continuous action spaces and has been applied to stock trading (Xiong et al., 2018), commodity futures (Liang et al., 2018), and multi-asset allocation (Soleymani and Paquet, 2021). Soft Actor-Critic (SAC) (Haarnoja et al., 2018), which maximizes a combination of expected return and entropy, has shown robustness advantages in financial settings due to its inherent exploration mechanism.

A common thread across these studies is that they operate in an *online* setting, where the agent simultaneously interacts with the market environment and updates its policy. This conflation of training and evaluation creates a risk of look-ahead bias, particularly in simulation-based studies where the environment is replayed from historical data and the agent can, in principle, extract information about future returns from the reward signal during training.

2.3 Offline Reinforcement Learning and Conservative Q-Learning

Offline RL—also known as batch RL—addresses the look-ahead bias problem by restricting training to a fixed dataset collected by a *behavioral policy* that is separate from the policy being learned (Levine et al., 2020). The agent never interacts with the live environment during training, eliminating any possibility of exploiting future information. This paradigm is particularly attractive for finance, where live experimentation is costly and historical data is abundant.

The central challenge of offline RL is the *distributional shift* problem: Q-values for actions that are underrepresented in the offline dataset can be severely overestimated, leading to policies that select actions the behavioral policy would never take and that may perform catastrophically in deployment. Kumar et al. (2020) propose Conservative Q-Learning (CQL) to address this. CQL augments the standard Bellman loss with a regularization term that penalizes the expected Q-value of actions sampled from the current policy while rewarding the expected Q-value of actions sampled from the behavioral data. Formally, this conservatism penalty minimizes a lower bound on the true Q-value for out-of-distribution actions, preventing the agent from assigning unrealistically high values to unexplored regions of the action space. Kumar et al. (2020) demonstrate that CQL achieves state-of-the-art performance on the D4RL offline RL benchmark (Fu et al., 2020), outperforming prior methods including Behavior-Cloning, BCQ (Fujimoto et al., 2019), and BEAR (Kumar et al., 2019).

In finance, offline RL has attracted growing attention. Yuan et al. (2025) apply an offline policy within a bilevel optimization framework for sequential portfolio optimization, highlighting the practical importance of ensuring that the offline policy does not merely memorize the optimal actions from the training data. The present paper is, to

our knowledge, among the first to apply CQL specifically to multi-asset equity portfolio optimization in the Brazilian market, combining offline safety guarantees with explicit tail-risk control.

2.4 CVaR in Portfolio Optimization

Conditional Value-at-Risk (CVaR), also known as Expected Shortfall or Tail Value-at-Risk, was formally introduced as a coherent risk measure by Rockafellar and Uryasev (2000), who also proposed a linear-programming formulation that makes CVaR minimization tractable in the context of portfolio optimization. Unlike Value-at-Risk (VaR), which only characterizes the loss threshold at a given confidence level but ignores the severity of losses beyond that threshold, CVaR measures the expected loss conditional on exceeding the VaR threshold. CVaR satisfies the axioms of coherent risk measures—translation invariance, monotonicity, positive homogeneity, and subadditivity—making it a theoretically sound basis for portfolio risk control (Artzner et al., 1999).

The application of CVaR to reinforcement learning reward design has received increasing attention in the risk-sensitive RL literature. Tamar et al. (2015) develop a policy-gradient algorithm for CVaR-constrained MDPs, proving convergence to a local optimum. More recently, Dabney et al. (2018) propose Distributional RL methods that learn the full return distribution rather than only its expectation, enabling quantile-based risk measures to be incorporated directly into the value function. In the portfolio context, a CVaR reward penalizes the agent for generating return distributions with heavy left tails, which is particularly relevant in emerging markets where political risk, currency crises, and commodity shocks produce return distributions with excess kurtosis and negative skewness.

2.5 Macroeconomic Regimes and Emerging Markets

The literature on regime-switching models in finance dates to Hamilton (1989), who demonstrated that U.S. GDP growth follows a two-state Markov switching process. Subsequent work extended regime-switching to equity returns, showing that volatility, correlation structure, and expected returns vary substantially across expansionary and recessionary regimes (Ang and Bekaert, 2002). In portfolio management, conditioning on regimes has been shown to improve risk-adjusted performance: Guidolin and Timmermann (2007) find that mean-variance investors who condition on business-cycle regimes achieve economically large utility gains relative to unconditional allocations.

The Brazilian market presents a particularly compelling case for regime-aware strategies. The SELIC rate—the primary instrument of Brazilian monetary policy—has fluctuated between 2% and over 14% during our sample period (2015–2024), with each major cycle coinciding with distinct equity-market dynamics. Sachsida and Mendonça (2006)

document that Brazilian asset prices are more responsive to domestic monetary shocks than to international ones, underscoring the importance of incorporating SELIC dynamics into portfolio models. Additionally, the Brazilian market is characterized by relatively high equity premiums, elevated credit spreads, and persistent currency volatility that amplify the macro-financial transmission mechanism compared with developed markets (Bonomo et al., 2011). These structural features motivate conditioning the allocation on the monetary stance rather than on price history alone. They also set a ceiling on what the state vector used here can capture: sovereign credit risk and implied volatility are distinct dimensions of emerging-market risk, and neither is observed in this study (Section 3.2).

3 Data

3.1 Asset Universe

The equity universe consists of ten large-cap Brazilian stocks listed on the B3 exchange: PETR4, VALE3, ITUB4, BBDC4, ABEV3, WEGE3, RENT3, SUZB3, B3SA3, and BBAS3. These ten tickers were selected based on three criteria: (i) continuous listing throughout the full sample period January 2015 to December 2024, (ii) membership in the Ibovespa index for at least eight of the ten years, and (iii) average daily traded volume exceeding R\$ 100 million, ensuring that the portfolio weights implied by the optimization are implementable without significant market-impact costs. The ten assets span six sectors—energy and materials (PETR4, VALE3, SUZB3), financials (ITUB4, BBDC4, BBAS3, B3SA3), consumer staples (ABEV3), industrials (WEGE3), and services (RENT3)—providing cross-sector diversification within the concentrated structure of the Brazilian equity market.

Daily adjusted closing prices were obtained from Yahoo Finance via the `yfinance` library. Log returns were computed as $r_{i,t} = \log(P_{i,t}/P_{i,t-1})$ for each asset i and trading day t . The full sample spans January 2015 to December 2024, comprising 2,487 trading days. Table 1 reports summary statistics for log returns over the full period. The cross-sectional dispersion in annualized returns is substantial: WEGE3 delivers the highest annualized return at 23.9%, followed by PETR4 at 26.0%, while ABEV3 records the lowest at 1.2%, reflecting both sector dynamics and firm-specific factors. The equal-weight portfolio achieves a mean daily return of 0.000545 (annualized: $\approx 13.7\%$) with a daily standard deviation of 0.015521 and a $\text{CVaR}_{0.95}$ of -0.0249 .

3.2 Macroeconomic Variables

Two macroeconomic series enter the state vector, both retrieved from the Banco Central do Brasil public time-series API (SGS, `api.bcb.gov.br`). A third series, IPCA inflation

(SGS 433), is collected and stored but is not used by any model reported here.

SELIC target rate (SGS 432). The policy rate set by the Monetary Policy Committee (COPOM), in percent per annum. This is the target rate rather than the realised overnight rate (SGS 11): the target is the instrument the committee actually decides on, is piecewise constant between meetings, and therefore gives an unambiguous reading of the policy stance on any given day. It enters the state vector as a rolling 252-day z-score, which removes the level drift and leaves the variable comparable across the sample.

Bank lending spread (SGS 20786). The average spread on non-earmarked credit operations to households, in percentage points, that is, the difference between the average rate charged by banks on such operations and their cost of funding. It enters the state vector as a rolling 252-day z-score. This is a domestic bank-credit measure, not a corporate-bond-versus-government spread, and it captures tightening in the transmission of monetary policy to household credit rather than sovereign or corporate default risk.

We note explicitly what is *not* in the state vector, because both are natural candidates that a reader may expect: there is no sovereign CDS series and no implied-volatility index. Neither was available to this study at daily frequency over the full sample without a paid data subscription, and no proxy for either is used. The macro block of the state vector is therefore two scalars plus the regime encoding of Section 3.3, and any claim this paper makes about “macro conditioning” refers to that and nothing more.

Both series are aligned to the equity trading calendar by forward fill, since neither reports on every trading day.

3.3 SELIC Regime Classification

Monetary-policy regimes are identified from the direction of the SELIC target using a rule-based classification over a trailing quarter. Let $\Delta_{63}(t)$ be the change in the SELIC target over the previous 63 trading days. The regime at date t is labeled *Rising* if $\Delta_{63}(t) > +0.25$ percentage points, *Falling* if $\Delta_{63}(t) < -0.25$, and *Stable* otherwise. The quarter-long window rather than the single most recent COPOM decision is deliberate: the committee meets eight times a year and often moves in a sequence of same-signed steps, so a one-meeting rule would flip the label during pauses inside an otherwise continuous cycle. The ± 0.25 threshold is one standard policy increment, which keeps a single minimum-size move from being read as a change of regime. The classification is strictly backward-looking—only information available at each date enters it—so it introduces no look-ahead bias. Table 2 summarizes the regime distribution across the 2,487-day sample.

Over the full period, the Stable regime accounts for the majority of trading days (42.1%, or 1,048 days), reflecting the extended periods between rate decisions. The Rising regime covers 23.5% of the sample (584 days), concentrated primarily in 2015–2016 and 2021–2023. The Falling regime accounts for the remaining 34.4% (855 days), correspond-

Table 1: Descriptive Statistics of B3 Assets (2015–2024)

Ticker	Mean (ann.)	Std (ann.)	Skewness	Kurtosis	JB p-value	CVaR _{0.95}	Max Drawdown
ABEV3	0.0124	0.2640	-0.4348	9.4934	0.0000	-0.0372	-0.5695
B3SA3	0.1568	0.3747	-0.1092	4.5755	0.0000	-0.0512	-0.6409
BBAS3	0.1450	0.4061	-0.5782	10.8046	0.0000	-0.0572	-0.6272
BBDC4	0.0347	0.3545	-0.5171	8.9735	0.0000	-0.0511	-0.6345
ITUB4	0.1035	0.3101	-0.4309	7.2240	0.0000	-0.0437	-0.5040
PETR4	0.2602	0.4784	-1.0948	13.8397	0.0000	-0.0709	-0.7440
RENT3	0.1318	0.4196	-0.2524	10.8258	0.0000	-0.0582	-0.6604
SUZB3	0.1310	0.3272	0.2335	10.6168	0.0000	-0.0469	-0.6130
VALE3	0.1596	0.4360	-0.2748	8.4425	0.0000	-0.0611	-0.7135
WEGE3	0.2391	0.3350	-0.4542	10.1939	0.0000	-0.0451	-0.5361

Table 2: Equal-Weight Portfolio Statistics by SELIC Regime (2015–2024)

Regime	N Days	Mean Return (daily)	Std (daily)	Sharpe (ann.)	CVaR _{0.95}	Avg SELIC (%)
Rising	584	-0.0002	0.0130	-0.2272	-0.0272	10.2230
Stable	1047	0.0008	0.0153	0.8781	-0.0329	9.9833
Falling	855	0.0007	0.0173	0.6199	-0.0400	8.3272

ing to the easing cycle of 2017–2020 and the initial phase of the 2023–2024 normalization.

The economic significance of these regimes is confirmed by the large variation in aggregate Sharpe ratios: the equal-weight portfolio achieves a Sharpe ratio of -0.23 during Rising-rate periods, $+0.88$ during Stable periods, and $+0.62$ during Falling periods. This three-way split provides a natural test of whether a regime-aware portfolio strategy can exploit this heterogeneity to improve risk-adjusted performance.

3.4 Stationarity

All ten log-return series were tested for stationarity using the Augmented Dickey-Fuller (ADF) test. All ten tickers reject the unit-root null hypothesis at the 1% significance level ($p < 0.001$), confirming that log returns are stationary in levels. This is consistent with standard results in the empirical finance literature and validates the use of log returns as the primary input to the model without additional differencing.

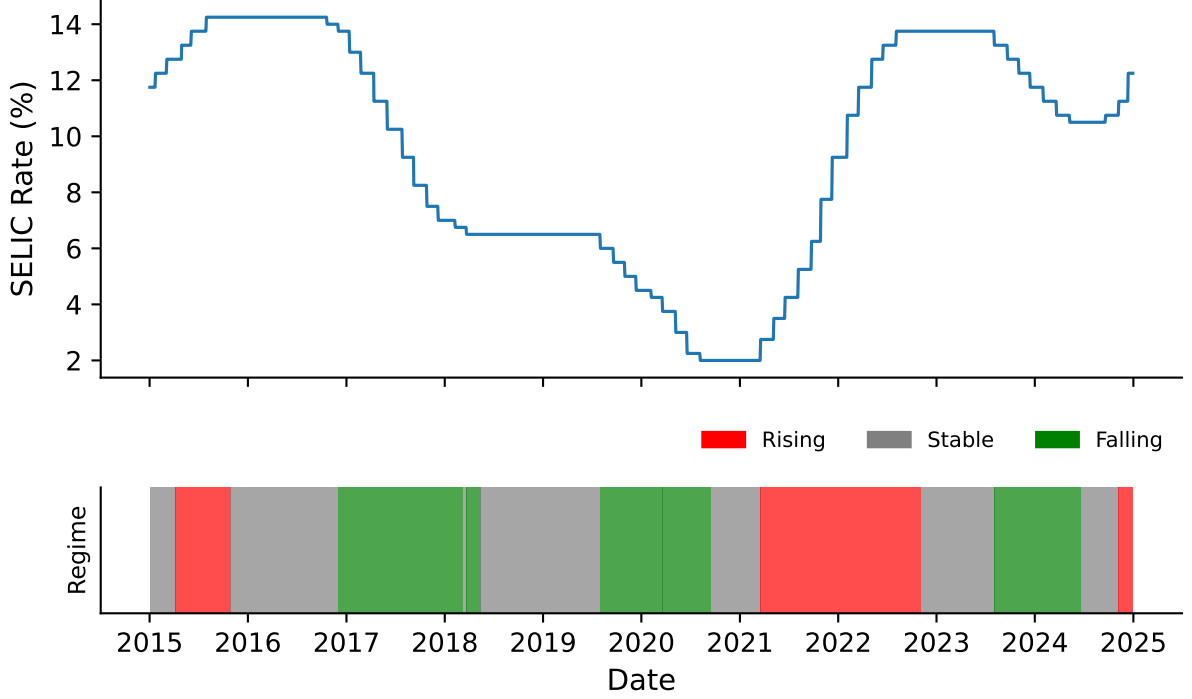


Figure 1: SELIC overnight rate and identified monetary-policy regimes, 2015–2024. Shaded bands indicate Rising (red), Stable (gray), and Falling (blue) rate regimes classified from COPOM decisions.

4 Methodology

4.1 Markov Decision Process Formulation

Portfolio rebalancing is cast as a discrete-time Markov Decision Process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{R}$ is the reward function, $\mathcal{P}$ is the transition kernel, and $\gamma \in (0, 1)$ is the discount factor.

State space. At each rebalancing date t , the state $s_t \in \mathcal{S} \subset \mathbb{R}^{225}$ is a concatenation of five groups of features:

1. *Normalized log returns* $\tilde{r}_{i,t-k}$ for $i = 1, \dots, 10$ and $k = 0, \dots, 19$ (lookback $T = 20$ days), standardized to zero mean and unit variance over a rolling window: 200 dimensions.
2. *Realized volatility* $\hat{\sigma}_{i,t}$ estimated over the same 20-day window for each asset: 10 dimensions.
3. *Momentum* $m_{i,t} = \sum_{k=1}^{20} r_{i,t-k}$ for each asset: 10 dimensions.
4. *Macroeconomic scalars*: the SELIC target (1 dimension) and the bank lending spread (1 dimension), each as a rolling 252-day z-score (Section 3.2).

5. *SELIC regime one-hot encoding*: three binary indicators (Rising, Stable, Falling): 3 dimensions.

The total state dimension is $200 + 10 + 10 + 1 + 1 + 3 = 225$.

Action space. The action $a_t \in \mathcal{A}$ is a portfolio weight vector $\mathbf{w}_t = (w_{1,t}, \dots, w_{N,t})$ constrained to the N -dimensional unit simplex:

$$\mathcal{A} = \Delta^N = \left\{ \mathbf{w} \in \mathbb{R}^N : w_i \geq 0, \sum_{i=1}^N w_i = 1 \right\}, \quad (1)$$

with $N = 10$. This formulation enforces long-only positions and full investment, consistent with the regulatory constraints applicable to many Brazilian fund structures.

Reward function. The reward at time t combines a tail-risk penalty with a linear expected-return component:

$$r_t = -\text{CVaR}_\alpha(\mathbf{R}_{p,t}) + \lambda \mathbb{E}[\mathbf{R}_{p,t}] \quad (2)$$

where $\mathbf{R}_{p,t} = \mathbf{w}_t^\top \mathbf{r}_t$ is the portfolio return, $\alpha = 0.95$ is the CVaR confidence level, and $\lambda = 0.5$ controls the trade-off between downside-risk minimization and expected-return maximization. Negative CVaR values indicate severe drawdowns, so the minus sign in Equation (2) converts the risk penalty into a reward to be maximized.

The CVaR at confidence level α is defined as:

$$\text{CVaR}_\alpha(X) = -\frac{1}{1-\alpha} \int_{-\infty}^{\text{VaR}_\alpha(X)} x dF_X(x) \quad (3)$$

where $\text{VaR}_\alpha(X) = \inf\{x : F_X(x) \geq \alpha\}$ is the Value-at-Risk threshold and F_X is the cumulative distribution function of X . Operationally, CVaR is estimated as the average of the $(1-\alpha)$ worst returns obtained by applying the candidate weights $\mathbf{w}_t$ to the trailing window $\mathbf{r}_{t-T:t}$, with $T = 20$ days.

This deserves emphasis, because it is a property of the implementation rather than of the CVaR objective in general. The window scored by the reward is the same window of returns that the state already encodes in its first 200 dimensions. The reward is consequently a deterministic function of the state and the action: it introduces no information beyond what the agent observes, and the transition carries no feedback about the consequence of the action taken. The Sharpe-based reward used in configurations A1 and A2, by contrast, is the realized portfolio return on day t under the behavioral action, which is not contained in the state. Section 6.1 returns to this asymmetry, which we regard as the central methodological finding of the ablation.

The discount factor is set to $\gamma = 0.99$, reflecting a preference for near-term rewards while still valuing long-horizon portfolio performance.

4.2 Conservative Q-Learning

Standard Q-learning applied to offline data suffers from distributional shift: the Bellman operator may assign high Q-values to actions that are absent from the offline dataset and that would perform poorly in deployment. CQL (Kumar et al., 2020) addresses this by augmenting the standard SAC loss with a conservatism penalty that pushes down Q-values for out-of-distribution actions while pulling up Q-values for actions observed in the behavioral data. The CQL loss is:

$$\mathcal{L}_{\text{CQL}}(\theta) = \mathcal{L}_{\text{SAC}}(\theta) + \alpha_{\text{CQL}} \cdot \mathbb{E}_{s \sim \mathcal{D}} \left[\log \sum_a \exp Q_\theta(s, a) - \mathbb{E}_{a \sim \beta(s)}[Q_\theta(s, a)] \right] \quad (4)$$

where $\mathcal{L}_{\text{SAC}}(\theta)$ is the standard Soft Actor-Critic Bellman residual loss, $\mathcal{D}$ is the offline dataset, $\beta(s)$ is the behavioral policy distribution at state s , and $\alpha_{\text{CQL}} > 0$ is the conservatism coefficient (distinct from the CVaR confidence level α).

The first term inside the brackets, $\log \sum_a \exp Q_\theta(s, a)$, is the log-sum-exp over the current policy’s action distribution and approximates the maximum Q-value; minimizing it discourages the agent from assigning high value to actions not grounded in the behavioral data. The second term, $\mathbb{E}_{a \sim \beta(s)}[Q_\theta(s, a)]$, rewards accurate value estimation for actions that were actually taken by the behavioral policy, preventing the conservatism penalty from depressing Q-values uniformly.

In financial applications, the conservatism property of CQL is particularly valuable: it prevents the agent from constructing highly concentrated or speculative portfolios that would only appear optimal by exploiting spurious correlations in a finite training sample. This implicit regularization is analogous to the shrinkage estimators commonly used in classical mean-variance optimization but is learned end-to-end through the offline RL training procedure.

The CQL agent is implemented using the `d3rlpy` library (Seno and Imai, 2022) with a two-layer actor network of hidden dimension 256 and a corresponding critic. Portfolio weights are produced by a softmax output activation that automatically enforces the simplex constraint in Equation (1). The hyperparameters α_{CQL} and λ are selected via cross-validation on a held-out validation fold within the training set.

4.3 Walk-Forward Evaluation Protocol

Two evaluation protocols are used in this paper, and they are not the same protocol applied to different strategies. The distinction governs how the comparison in Section 5 should be read, so it is stated explicitly here.

Benchmarks: rolling walk-forward. The three classical strategies are evaluated with a rolling walk-forward scheme over the full 2,487-day sample, partitioned into 35

overlapping folds. Each fold consists of a 252-trading-day estimation window followed by an immediately subsequent 63-trading-day test window, corresponding to approximately one year of estimation data and one quarter of test data. Windows advance in 63-day increments, so successive estimation sets overlap but the test windows are non-overlapping and contiguous. Weights are re-estimated at every fold from the trailing estimation window only. Concatenating the 35 test windows yields a 2,205-day out-of-sample return series covering 2016–2024.

CQL agent: single chronological holdout. The offline RL agent is *not* retrained fold by fold. The transition dataset is split once in calendar time: transitions from 2016-01-13 to 2021-12-30 ($N = 1,485$) form the offline training set $\mathcal{D}$, generated by the MVP behavioral policy, and transitions from 2022-01-03 to 2024-12-27 ($N_{\text{eval}} = 748$) form a strict holdout that is touched neither during training nor during hyperparameter selection. The agent is trained once on $\mathcal{D}$ for a fixed number of gradient steps and then deployed frozen over the holdout; no weight update occurs after the calendar cutoff.

Both protocols guarantee that estimation precedes evaluation in calendar time, so neither admits look-ahead bias. They are not, however, equally favourable to the strategy being measured. The benchmarks re-estimate their parameters every quarter and therefore adapt to the evaluation period as it unfolds, whereas the agent’s policy is fixed at the end of 2021 and must generalise across three subsequent years of unseen regimes. Any comparison between the two is conservative with respect to the agent in this specific sense, and a per-fold retrained agent is the like-for-like counterpart that this study does not provide.

All comparisons reported in Section 5 are computed on the intersection of the two evaluation windows, that is, on the trading days for which the agent and the benchmarks both produce a return. The benchmark series ends on 2024-11-13, because the walk-forward requires a full 63-day test window and the sample does not accommodate a further fold; the agent’s series runs to 2024-12-27. Restricting to the common window removes 28 trading days from the agent’s series and is what makes the two sets of numbers comparable at all.

4.4 Benchmark Strategies

Three classical benchmark strategies are evaluated under the rolling walk-forward protocol described above:

Equal-Weight (EW). Portfolio weights are fixed at $w_i = 1/N = 0.1$ for all assets at every rebalancing date. This naive diversification rule (DeMiguel et al., 2007) requires no estimation and serves as a baseline that is robust to estimation error.

Minimum-Variance Portfolio (MVP). Weights are obtained by solving $\min_{\mathbf{w} \in \Delta^N} \mathbf{w}^\top \hat{\Sigma} \mathbf{w}$, where $\hat{\Sigma}$ is the sample covariance matrix estimated on the trailing 252-day training win-

dow. The MVP is the classical risk-minimization benchmark and is used as the behavioral policy to generate the offline dataset.

Risk Parity (RP). Weights are set so that each asset contributes equally to total portfolio variance. Risk contributions are equalized iteratively using the algorithm of Maillard et al. (2010).

All three benchmarks are rebalanced quarterly, at the fold boundaries of the walk-forward, and transaction costs are excluded from the primary analysis to ensure comparability across methods.

4.5 Ablation Study Design

To decompose the sources of performance of the full agent, we evaluate four configurations that differ by exactly one component each. All four are trained offline, on the same transition dataset, for the same 5,000 gradient steps, with the same seed; nothing but the named component changes between consecutive rungs.

- **A1 – CQL with conservatism disabled, Sharpe reward, no macro state.** The conservative weight is set to $\alpha_{\text{CQL}} = 0.001$, which reduces the CQL objective in Equation (4) to approximately its Soft Actor-Critic component: the penalty on out-of-distribution actions is present in form but not in magnitude. The reward is the realised daily portfolio return of the behavioral action, and the state vector excludes the five macro dimensions. This is the least constrained agent in the ladder, and it stands in for what an offline actor-critic without a conservatism term produces on this dataset.
- **A2 – CQL with conservatism, Sharpe reward, no macro state.** Identical to A1 except that the conservative weight is set to the value selected by the grid search of Section 4.2. Comparing A2 with A1 isolates the contribution of the conservatism penalty, not of the offline paradigm: both agents are offline.
- **A3 – CQL with conservatism, CVaR reward, no macro state.** Identical to A2 except for the reward, which becomes the CVaR-based objective of Equation (2). Comparing A3 with A2 isolates tail-risk-aware reward shaping.
- **A4 – CQL with conservatism, CVaR reward, with macro state.** The full proposed model, adding the SELIC one-hot encoding, the SELIC target level, and the bank lending spread to the state vector. Comparing A4 with A3 isolates the incremental contribution of the macro variables.

The design therefore attributes performance differences to the conservatism penalty, the reward function, and the macro state representation independently. It does *not* speak

to the online-versus-offline question: no online agent is trained in this study, and any claim about the offline paradigm would require one.

5 Results

5.1 Main Out-of-Sample Results

Table 3 reports out-of-sample performance for every strategy on the common evaluation window described in Section 4.3: the 720 trading days between 2022-01-03 and 2024-11-13 on which the agent and all three benchmarks produce a return. The four agent configurations are trained under an identical protocol (5,000 gradient steps, seed 42) and differ only in the component named in Section 4.5. The metrics reported are annualized return, Sharpe ratio, Sortino ratio, $\text{CVaR}_{0.95}$ (defined as the average loss in the worst 5% of daily return observations, reported as a positive number for clarity), maximum drawdown (MaxDD), and Calmar ratio (annualized return divided by absolute maximum drawdown). Every figure comes from a single evaluation window and a single seed and carries no interval estimate; Section 6.3 sets out what that precludes.

Among the three classical benchmarks, Equal-Weight delivers the highest Sharpe ratio (0.6076) and annualized return (10.34%), confirming the $1/N$ puzzle documented by DeMiguel et al. (2007) in the context of B3 equities. The Minimum-Variance Portfolio achieves the lowest $\text{CVaR}_{0.95}$ (0.0198) and the smallest maximum drawdown (-18.22%) but at the cost of markedly lower returns (5.16% annualized), yielding a Sharpe ratio of 0.3400. Risk Parity occupies an intermediate position, with a Sharpe ratio of 0.5561 and an annualized return of 8.94%.

The full CQL agent (configuration A4) achieves a Sharpe ratio of 0.4058 and an annualized return of 7.88%, with a $\text{CVaR}_{0.95}$ of 0.0260 and a maximum drawdown of -27.22% . This is -0.2018 in Sharpe relative to Equal-Weight and $+0.0657$ relative to MVP, the strategy that generated the behavioral data. Equal-Weight dominates A4 on every metric in Table 3: higher return, higher Sharpe, higher Sortino, lower tail loss, shallower drawdown and higher Calmar. Risk Parity likewise dominates it. Against MVP the picture is mixed rather than favourable: A4 earns more and has a higher Sharpe, but its tail loss is a third larger (0.0260 against 0.0198) and its drawdown half again as deep.

Two features of the table deserve to be stated plainly, because the rest of this section is an attempt to account for them. First, the proposed configuration is not the best agent in its own ablation: A2, which uses neither the CVaR reward nor the macro state, attains a Sharpe of 0.4464 against A4’s 0.4058. Second, $\text{CVaR}_{0.95}$ increases monotonically as the CVaR reward is introduced and then extended: 0.0222 for A1, 0.0221 for A2, 0.0241 for A3 and 0.0260 for A4. The two configurations that optimize a tail-risk objective have the worst realised tail loss of the seven strategies compared.

Table 3: Out-of-sample performance of all strategies, common evaluation window 2022-01-03 to 2024-11-13 (720 trading days). All configurations are trained under the identical protocol (5,000 gradient steps, seed 42). A4 is the full proposal (CQL + CVaR + macro state).

	Ann.Return	Ann.Vol	Sharpe	Sortino	CVaR _{0.95}	MaxDD	Calm
A1 (Unconstrained+Sharpe)	0.0262	0.1683	0.1557	0.2511	0.0222	-0.2314	0.11
A2 (CQL+Sharpe)	0.0802	0.1796	0.4464	0.7784	0.0221	-0.2491	0.32
A3 (CQL+CVaR)	0.0391	0.1851	0.2111	0.3500	0.0241	-0.2606	0.15
A4 (CQL+CVaR+Macro)	0.0788	0.1942	0.4058	0.6482	0.0260	-0.2722	0.28
1/N	0.1034	0.1701	0.6076	0.9808	0.0226	-0.1782	0.58
Min. Variance	0.0516	0.1516	0.3400	0.5545	0.0198	-0.1822	0.28
Risk Parity	0.0894	0.1607	0.5561	0.9013	0.0211	-0.1806	0.49

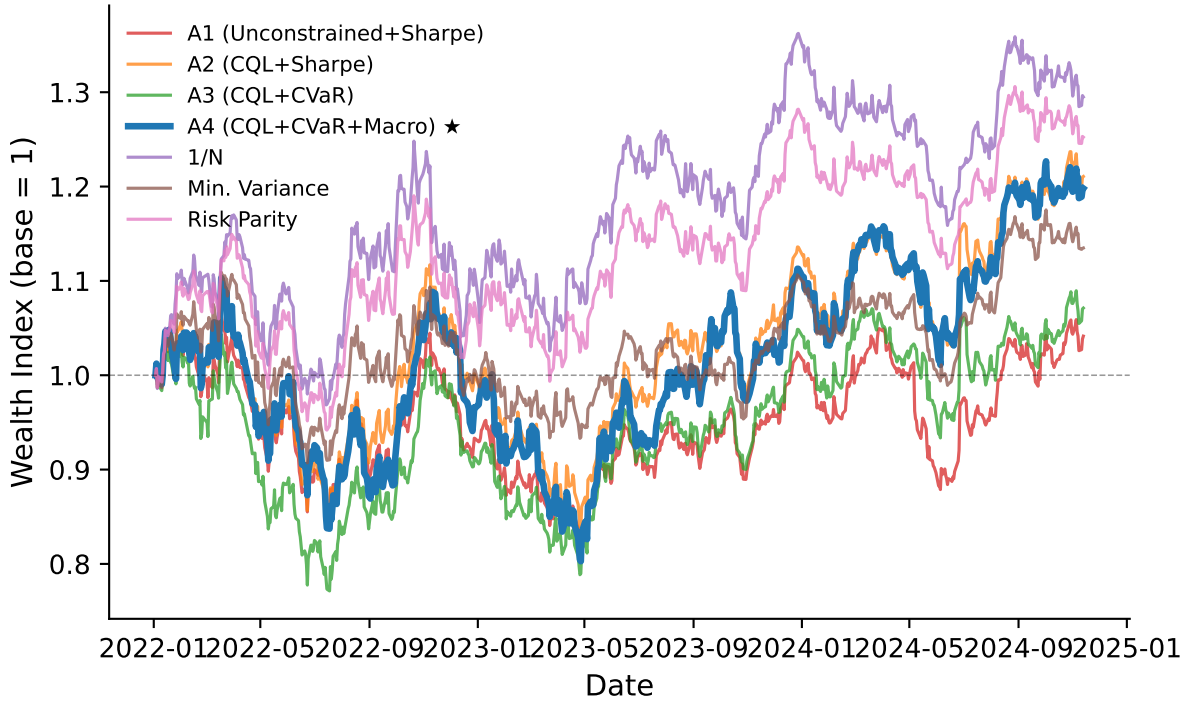


Figure 2: Cumulative wealth index for all strategies, out-of-sample period 2022–2024 (base = 1 at start of evaluation). A4 (CQL+CVaR+Macro) is plotted with a thicker line.

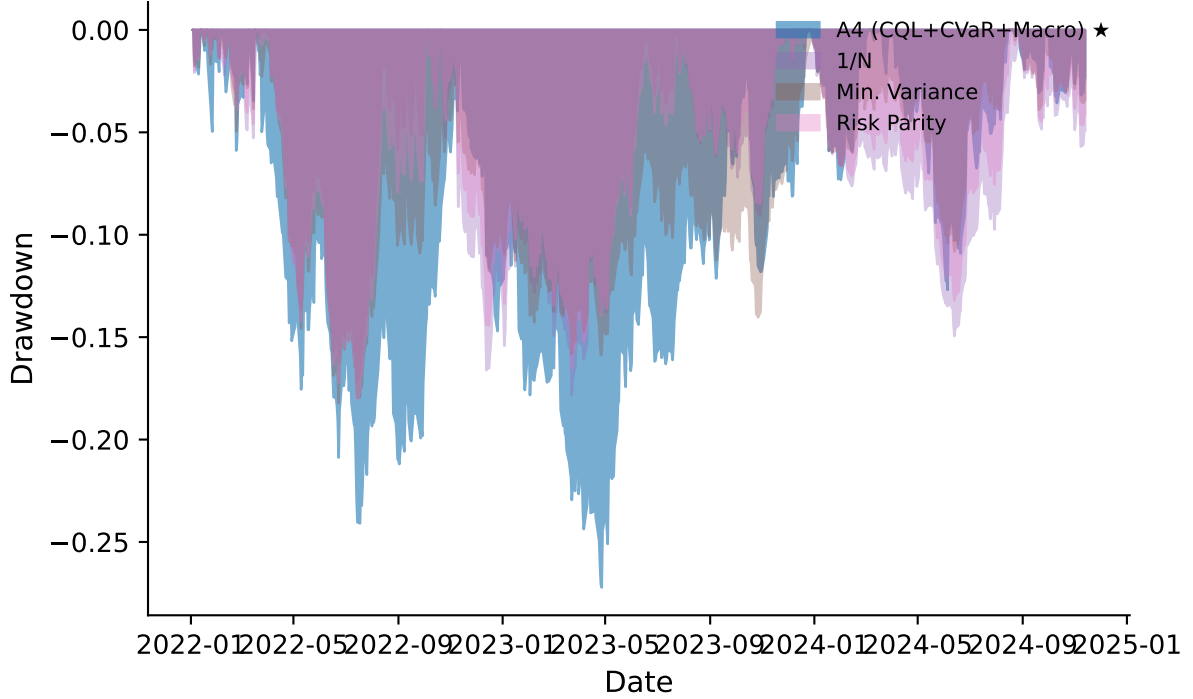


Figure 3: Drawdown trajectories for A4, Equal-Weight, MVP, and Risk Parity, out-of-sample period 2022–2024.

5.2 Sub-Period Analysis

We divide the out-of-sample period into three sub-periods to assess performance stability across distinct macroeconomic regimes:

2022 tightening phase. This period corresponds to the most aggressive phase of the SELIC hiking cycle, during which the rate rose from 9.25% to 13.75%. The agent does not provide downside protection here. A4 returns -1.94% annualized with a Sharpe of -0.086 and a $\text{CVaR}_{0.95}$ of 0.0297 , against $+11.89\%$, 0.573 and 0.0265 for Equal-Weight. This is the sub-period in which the motivating hypothesis predicted the clearest advantage, and it is the sub-period in which the agent performs worst in both return and tail loss.

2023 peak rate period. With the SELIC at its ceiling of 13.75% , A4 generates a Sharpe of 0.8374 against 1.3501 for Equal-Weight, and a larger tail loss (0.0243 against 0.0200). The agent is profitable in absolute terms and still behind the naive benchmark.

2024 easing phase. This is the one sub-period in which A4 wins, and it wins decisively: 9.50% annualized with a Sharpe of 0.633 , against -5.05% and -0.424 for Equal-Weight. Taken together with the two preceding sub-periods, the pattern is not one of regime intelligence but of persistent defensive positioning: the agent gives up return when equities rise and preserves capital when they fall. A lower-beta portfolio would produce the same three-period signature without any learning at all, and this study does not include the beta-matched comparison that would separate the two explanations.

Table 4: Regime-conditional performance by SELIC cycle phase.

Regime	Strategy	Sharpe	CVaR _{0.95}	Ann.Return
Falling	A4 (CQL+CVaR+Macro)	0.4609	0.0215	0.0751
	1/N	-0.4524	0.0155	-0.0599
	Min. Variance	-0.2184	0.0167	-0.0261
Stable	A4 (CQL+CVaR+Macro)	0.4929	0.0249	0.0928
	1/N	0.7318	0.0236	0.1264
	Min. Variance	0.7000	0.0191	0.1001
Rising	A4 (CQL+CVaR+Macro)	0.2811	0.0290	0.0641
	1/N	1.2050	0.0244	0.2385
	Min. Variance	0.3537	0.0222	0.0663

5.3 Regime-Conditional Performance

A central hypothesis of this paper is that macro regime information improves performance specifically during adverse monetary-policy environments. To test this, we compute performance metrics conditional on the SELIC regime label at each trading day.

Table 4 reports Sharpe ratios and CVaR by regime. The hypothesis is not supported, and the ordering it predicts is reversed.

In the Rising-rate regime, A4 achieves a Sharpe of 0.2811 against 1.2050 for Equal-Weight: the tightening phase is where the agent trails the naive benchmark by the widest margin, not the narrowest. In the Stable regime the gap narrows but persists, 0.4929 against 0.7318. In the Falling regime the ranking inverts: A4 achieves 0.4609 while Equal-Weight returns -0.4524 and MVP -0.2184 , making the agent the only strategy of the three with a positive Sharpe while rates come down.

Note that the sign of the market’s own regime-conditional performance differs between the full sample and the evaluation window. Over 2015–2024 the equal-weight portfolio earns an annualized Sharpe of -0.23 in rising-rate phases (Table 2), which is the empirical regularity that motivated conditioning on the cycle in the first place. Within the 2022–2024 evaluation window that regularity does not hold: equities rose through the tightening phase and fell through the easing phase. An agent whose training data encodes the historical association therefore inherits a mapping that the test period does not honour, and the regime-conditional results are as much a statement about that instability as about the agent.

5.4 Ablation Study

Table 6 reports the four configurations A1–A4. Performance does not improve monotonically along the ladder, and the two components the paper set out to defend are the two that cost performance.

Table 5: Sub-period performance analysis (2022–2024).

Period	Strategy	Ann.Return	Ann.Vol	Sharpe	Sortino	CVaR _{0.01}
2022 (Tightening)	A4 (CQL+CVaR+Macro)	-0.0194	0.2255	-0.0860	-0.1401	0.0290
	1/N	0.1189	0.2073	0.5735	0.9077	0.0200
	Min. Variance	0.0100	0.1870	0.0536	0.0955	0.0230
2023 (Peak/Stable)	A4 (CQL+CVaR+Macro)	0.1633	0.1950	0.8374	1.4073	0.0240
	1/N	0.2255	0.1670	1.3501	2.4030	0.0200
	Min. Variance	0.1072	0.1438	0.7457	1.2656	0.0180
2024 (Easing)	A4 (CQL+CVaR+Macro)	0.0950	0.1501	0.6329	0.9982	0.0190
	1/N	-0.0505	0.1190	-0.4244	-0.7206	0.0140
	Min. Variance	0.0361	0.1099	0.3290	0.5004	0.0140

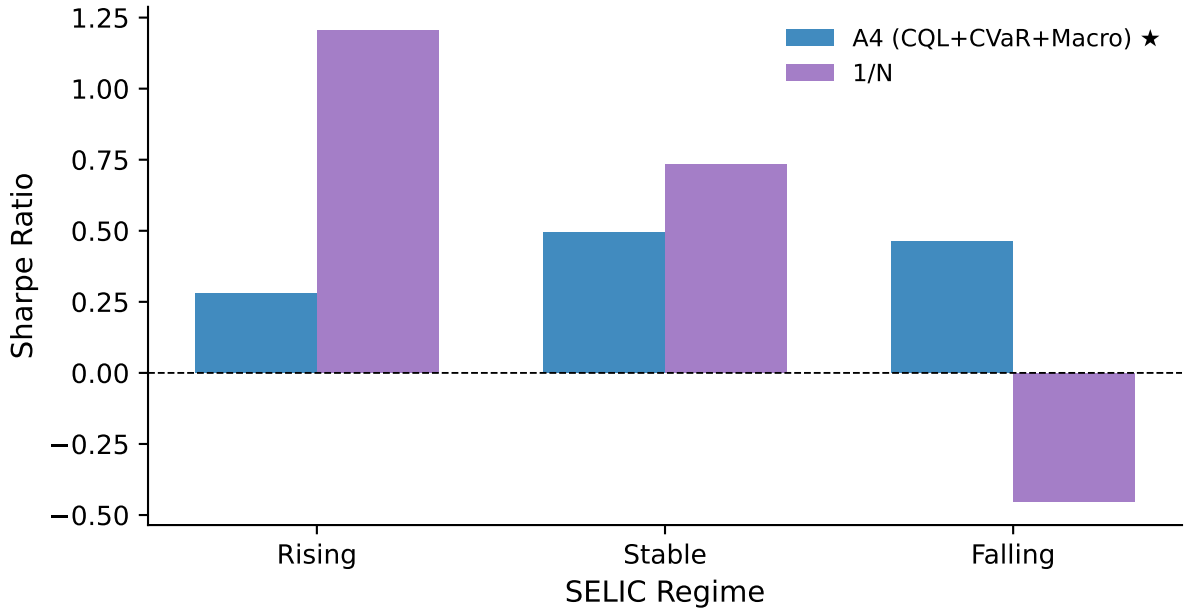


Figure 4: Regime-conditional Sharpe ratios for A4 and Equal-Weight across Rising, Stable, and Falling SELIC regimes.

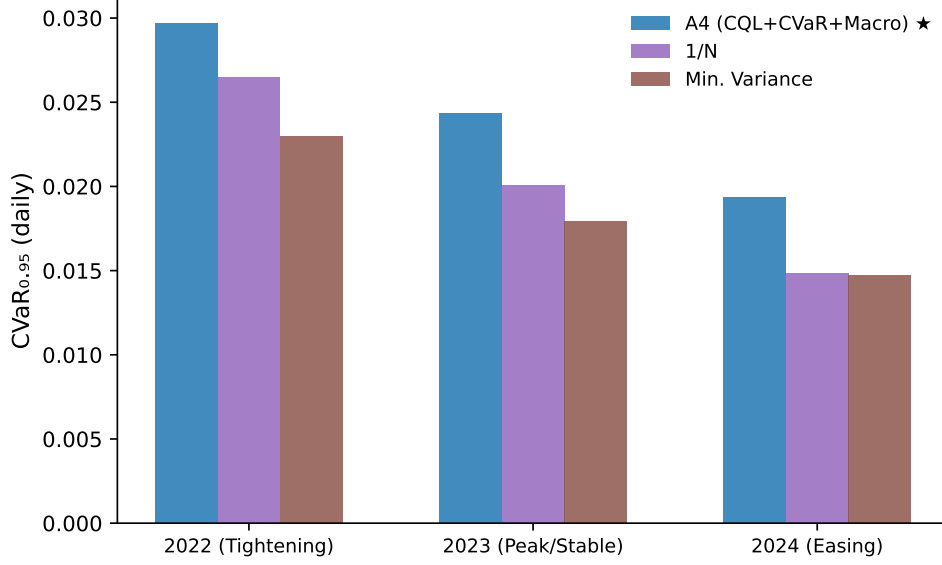


Figure 5: $\text{CVaR}_{0.95}$ by sub-period for A4, Equal-Weight, and MVP. Lower values indicate better tail-risk control.

A1 vs. A2: the conservatism penalty. Raising the conservative weight from 0.001 to the value selected by grid search lifts Sharpe from 0.1557 to 0.4464, leaves tail loss essentially unchanged (0.0222 to 0.0221) and increases maximum drawdown modestly (-23.14% to -25.49% before rounding). This is the largest single gain in the study and the only component that survives the ablation. It is also the result most consistent with the offline RL literature: penalising value estimates for actions absent from the behavioral data is what keeps an offline agent from extrapolating into regions the dataset cannot support. Note that this comparison says nothing about online versus offline training. Both configurations are offline; only the penalty differs.

A2 vs. A3: the CVaR reward. Replacing the realised-return reward with the CVaR objective lowers Sharpe from 0.4464 to 0.2111 and raises $\text{CVaR}_{0.95}$ from 0.0221 to 0.0241. The reward intended to control tail risk makes tail risk worse. Section 6.1 argues that this follows from how the signal is constructed rather than from any property of the market: the reward scores candidate weights against the same twenty-day window the state already encodes, so it carries no information the agent does not already hold and provides no feedback about the consequences of the action.

A3 vs. A4: the macro state. Adding the SELIC target, the lending spread and the regime encoding lifts Sharpe from 0.2111 to 0.4058 while raising $\text{CVaR}_{0.95}$ further, from 0.0241 to 0.0260, and deepening maximum drawdown to -27.22% . The macro block recovers roughly four fifths of the Sharpe lost to the reward change without recovering any of the tail deterioration. Two readings are available and the data does not separate them. The macro variables may carry genuine information about the allocation decision; or, because the reward supplies no feedback, the additional state dimensions may simply

Table 6: Ablation study and benchmark comparison, out-of-sample 2022–2024. A1: CQL with conservatism disabled + Sharpe reward, no macro. A2: CQL + Sharpe reward, no macro. A3: CQL + CVaR reward, no macro. A4: CQL + CVaR reward + macro state (full proposal).

Config	Ann.Return	Ann.Vol	Sharpe	Sortino	CVaR _{0.95}	MaxDD	Calmar
A1	0.0240	0.1680	0.1427	0.2287	0.0223	-0.2314	0.1036
A2	0.0711	0.1798	0.3957	0.6799	0.0224	-0.2491	0.2856
A3	0.0347	0.1848	0.1878	0.3091	0.0242	-0.2606	0.1331
A4	0.0671	0.1936	0.3468	0.5528	0.0260	-0.2722	0.2466
EqualWeight	0.1034	0.1701	0.6076	0.9808	0.0226	-0.1782	0.5800
MVP	0.0516	0.1516	0.3400	0.5545	0.0198	-0.1822	0.2829
RiskParity	0.0894	0.1607	0.5561	0.9013	0.0211	-0.1806	0.4950

be changing which trailing-window portfolio the agent converges to. Establishing the first reading requires re-running A3 and A4 under a reward that is not measurable from the state, which is the experiment Section 6.1 identifies as necessary.

The ladder as a whole. The best agent in the study is A2, not the proposed A4, and A2 uses neither of the two components this paper contributes. Every agent configuration is below Equal-Weight and below Risk Parity on Sharpe. The honest summary is that of the three ingredients tested, one helps, one hurts, and the third cannot be interpreted until the second is fixed.

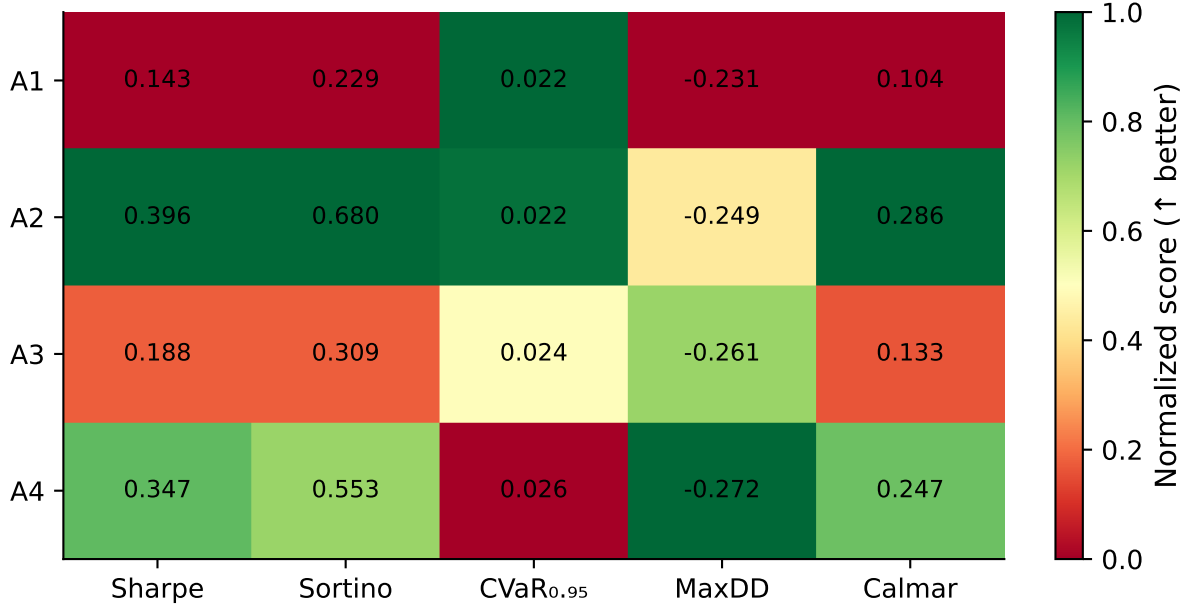


Figure 6: Ablation heatmap: normalized performance metrics for configurations A1–A4. Green (1.0) indicates the best value in each metric column; red (0.0) indicates the worst. Raw metric values are shown in each cell.

6 Discussion

6.1 Economic Interpretation of the CVaR Reward

The motivating argument for a CVaR objective is sound and we do not retract it. Brazilian equities exhibit pronounced negative skewness and excess kurtosis relative to developed-market benchmarks, driven by periodic currency crises, commodity price shocks, and fiscal uncertainty. The Sharpe ratio, treating upside and downside deviations symmetrically, values a large positive outlier as much as it penalizes an equal negative one, which is not how a risk-averse investor evaluates outcomes.

The implementation tested here, however, does not deliver that objective, and the ablation makes the failure visible rather than hiding it. Moving from A2 to A3—the single change being the reward—degrades both the efficiency and the tail behaviour of the resulting portfolios. This is not the direction the motivating argument predicts, and the explanation is in the construction of the reward rather than in the market.

Consider how the reward at date t is computed (Section 4.1). The portfolio return series entering the CVaR estimate is $\mathbf{r}_{t-T:t}^\top \mathbf{w}_t$ with $T = 20$: the candidate weights are scored against the *preceding* twenty trading days. The state at the same date, by construction, contains the normalized returns of those same twenty days. The reward is therefore a deterministic function of the state and the action, and contributes no information that the agent does not already observe. There is no environment feedback in the usual sense: maximizing this signal is not a decision problem about the future but a re-scoring of data the agent has already seen. Under this objective the optimal policy is the one that minimizes the trailing twenty-day tail loss of the observed window, which makes the agent a rolling short-window tail-risk optimizer. A twenty-day estimate of a 95% tail is built on one observation, and the out-of-sample behaviour of A3 is what one would expect of a portfolio rule fitted to that.

The contrast with A1 and A2 is instructive precisely because it is structural. Their reward is the realized portfolio return on day t under the behavioral action, a quantity that is absent from the state and only becomes known after the action is fixed. That is genuine feedback, and it is the configuration in which the conservatism penalty produces its clearest gain. The ablation, read this way, does not show that tail-risk-aware objectives fail in the Brazilian market. It shows that a reward which is measurable from the state alone cannot teach an agent anything, whatever quantity it is named after.

The correction is well defined and is the first item of future work: score the action against the returns that follow it, estimating the tail over a forward window $\mathbf{r}_{t:t+T}$, or over a longer trailing window that is deliberately excluded from the state. Until that experiment is run, no claim in this paper about CVaR reward shaping should be carried forward, in either direction. Note also that $\lambda = 0.5$ was fixed a priori and never tuned; the grid search of Section 4.2 varied only the conservative weight.

6.2 Role of Macroeconomic State Variables

Adding the macro block lifts Sharpe from 0.2111 to 0.4058 while leaving tail loss and drawdown worse than in either Sharpe-reward configuration. The gain is real in the arithmetic and difficult to attribute, for two reasons that compound.

The first is the reward. A3 and A4 share an objective that, as argued in Section 6.1, supplies no feedback: it is computable from the state. Adding five dimensions to a state under such an objective changes which trailing-window portfolio the optimizer settles on, and that change can raise or lower out-of-sample Sharpe without the new dimensions carrying information about the allocation decision at all. The comparison that would separate the two accounts, A3 against A4 under a reward scored on returns the agent has not yet seen, has not been run.

The second is the regime-conditional evidence, which points against the mechanism the hypothesis proposes. If the macro block were informative about tightening cycles specifically, the agent’s advantage should concentrate in the rising-rate regime. It is inverted: the agent trails the naive benchmark most widely precisely there (0.2811 against 1.2050) and leads only while rates fall (0.4609 against -0.4524). That pattern is what a persistently defensive allocation produces mechanically, and it does not require the agent to have learned anything about monetary policy. A beta-matched comparison would test this directly and is not part of this study.

One methodological point survives intact and is worth stating, because it is the part of the original argument that the data does not touch: both macro inputs are observable in real time and require no forecast or latent-state estimate, so an agent conditioned on them is implementable without a nowcasting apparatus. That property is a precondition for the approach, not evidence for it.

6.3 Limitations

Several limitations of this study should be acknowledged. The first three are, in our view, the ones that most constrain how the results of Section 5 may be read.

No measure of uncertainty. Every number reported in this paper comes from a single evaluation window and a single random seed. We provide no confidence interval on any Sharpe ratio and no test of the difference between any two strategies. With 720 trading days of out-of-sample data and a training procedure whose outcome depends on initialisation, differences of the magnitude reported here are well within the range that seed variation alone can produce in offline RL. The ordering of the configurations should therefore be read as a description of this run, not as an estimate of a population quantity. Repeating the ablation across seeds and reporting the dispersion is the single most valuable extension of this work, and it is not expensive: the whole ladder runs in under two hours on a laptop CPU.

Asymmetric evaluation protocols. As set out in Section 4.3, the benchmarks re-estimate their weights every quarter while the agent is trained once and then frozen for three years. The comparison is not like-for-like, and the direction of the bias is against the agent. A per-fold retrained agent would settle this, and this study does not provide one.

A thin macro block. The macro information available to the agent is two scalars and a three-way regime encoding (Section 3.2). Sovereign credit risk and implied volatility, the two variables most often cited as carrying emerging-market stress, are absent. Any conclusion about the value of “macro conditioning” is therefore a conclusion about this particular thin encoding of the policy stance, and says little about a richer one.

Transaction costs, and why they are decisive here. The evaluation protocol excludes transaction costs, and this is the assumption on which the reported comparison rests. The agent does not rebalance quarterly. The evaluation applies the policy’s predicted weights on every trading day, so the agent rebalances daily, while the benchmarks hold their weights fixed between quarterly re-estimations. Measuring one-way turnover as $\tau_t = \sum_i |w_{i,t} - w_{i,t-1}|$ by inference over the evaluation set gives annualised turnover of $38.5\times$ portfolio value for A1 and $87.9\times$ for A2, against approximately $1.0\times$ for the Minimum-Variance Portfolio, $0.5\times$ for Risk Parity, and zero for Equal-Weight, which never trades.

At a round-trip cost of 10 basis points—a conservative figure for liquid B3 names once brokerage, exchange fees, and slippage are included—this corresponds to roughly 3.9% to 8.8% per year for the agents and about 0.1% per year for the benchmarks. Annualised returns of the agents in the evaluation window are of the order of 9%. A cost differential of that size does not merely reduce the magnitude of any outperformance: it is of the same order as the entire return being compared. No conclusion in Section 5 should be read as a statement about implementable performance, and any future version of this study must either constrain turnover in the reward or report results net of a cost model.

A related observation concerns concentration. The mean maximum weight assigned by the full agent is 0.58, and on 525 of the 748 evaluation days a single asset exceeds half of the portfolio. The conservatism penalty of CQL bounds the agent to actions supported by the behavioral data; it does not, and is not designed to, impose diversification.

No short selling. The simplex constraint (Equation (1)) restricts the agent to long-only positions. This is consistent with regulatory constraints on many Brazilian retail and institutional funds but prevents the agent from hedging systematic risk by taking short positions in overvalued assets or index futures. Extending the framework to a box-constrained action space would be a natural next step.

Regime shift risk. The regime classification is based on COPOM decisions, which are discrete events occurring eight times per year. In periods of rapid or unprecedented monetary-policy changes—such as the emergency rate cuts during the COVID-

19 pandemic—the regime label may lag the true economic environment by several weeks. More sophisticated regime-detection methods, such as hidden Markov models or real-time macro nowcasting, could reduce this lag.

Asset universe concentration. The ten-asset universe, while representative of the B3 blue-chip segment, is highly concentrated in financials and commodity-linked sectors. Expanding the universe to include small- and mid-cap stocks, fixed-income instruments, or real-estate investment trusts (REITs) would test the generalizability of the proposed framework.

6.4 Practical Implications

We see no basis in these results for recommending the proposed strategy to a Brazilian fund manager, and three findings argue specifically against it. The agent is dominated by an equal-weight portfolio on every metric measured; it rebalances daily, at a turnover that would carry costs of the same order as its entire return; and it concentrates more than half of the portfolio in a single asset on 525 of 748 evaluation days. Any of the three would be disqualifying on its own.

What the study does offer a practitioner is narrower and, we think, more durable. The offline formulation itself holds up: training on a fixed historical dataset, with a conservatism penalty bounding the policy to actions the data supports, produced the single clearest improvement in the ablation and requires no market simulator whose realism would have to be defended to an auditor. That property makes the framework auditable in a way online RL in a simulated market is not, and it is independent of whether this particular agent works.

The finding most likely to save someone else’s time is the reward specification. An objective that can be computed from the observation is not a reward, however it is labelled, and in a financial MDP the mistake is easy to make: trailing-window risk statistics are natural quantities to optimize and natural quantities to put in a state vector, and putting the same window in both places is silent. Checking that the reward depends on data the agent has not yet observed costs nothing and would have caught this before the experiment ran.

7 Conclusion

This paper investigates whether an offline reinforcement learning agent equipped with a CVaR reward function and macroeconomic state variables can produce equity portfolios with superior risk-adjusted performance out-of-sample relative to classical benchmarks on the Brazilian stock exchange B3. We formulate the portfolio rebalancing problem as a Markov Decision Process, construct a 225-dimensional state vector that encodes asset-

level return dynamics together with SELIC regime indicators, the policy rate level and the bank lending spread, and train a Conservative Q-Learning (CQL) agent on an offline dataset generated by the Minimum-Variance Portfolio behavioral policy. Evaluated on 720 out-of-sample trading days shared with the benchmarks, the full proposed model is dominated by an equal-weight portfolio on every metric reported: Sharpe (0.4058 against 0.6076), Sortino, CVaR_{0.95}, maximum drawdown and Calmar. Risk Parity dominates it as well. The $1/N$ puzzle documented by DeMiguel et al. (2007) holds in this setting.

The value of the study lies in the ablation, which localizes the failure rather than reporting it in aggregate.

1. **Conservatism is the component that works.** Raising the CQL conservative weight from approximately zero to its tuned value is the single largest effect in the study, worth +0.29 in Sharpe, and it leaves tail loss unchanged. Bounding an offline policy to actions the behavioral data supports is doing real work on this dataset. We note that the ablation does not compare online with offline training: every configuration is offline, and no online agent was trained.
2. **The CVaR reward, as specified, is not a reward.** Replacing the realised-return signal with the CVaR objective costs -0.24 in Sharpe and, contrary to its purpose, worsens realised tail loss, which rises monotonically across the two configurations that use it. The cause is structural: the objective scores candidate weights against the same twenty-day window the state already encodes, making it a deterministic function of the observation. An agent maximising it receives no feedback about the consequences of its actions and converges to a rolling short-window tail optimizer fitted to a single tail observation. This is the finding we consider most useful to other practitioners, because the error is invisible at the level of the equations and appears only when the reward and the state are read side by side.
3. **Regime conditioning is unresolved, and the evidence available points the wrong way.** The macro block recovers +0.20 in Sharpe but not the tail deterioration, and its regime-conditional pattern inverts the motivating hypothesis: the agent trails the naive benchmark most widely during tightening, and leads only while rates fall. That signature is what a persistently defensive allocation produces without learning anything about monetary policy. Because A3 and A4 share the defective reward, the comparison between them cannot settle the question either way.

A fourth result is operational rather than statistical. The evaluation applies the policy daily, giving measured turnover of 38 to 88 times portfolio value per year against roughly one for the benchmarks. At ten basis points per round trip this is 4–9% annu-

ally, against agent returns of 2.6–8.0%. No comparison in this paper is informative about implementable performance, and we report none as such.

The order of future work follows from the above, and the first two items are prerequisites rather than extensions. First, the reward must be re-specified so that it scores an action against returns the agent has not observed, and A3 and A4 re-run; until then the paper’s second and third questions remain open rather than answered. Second, the ablation must be repeated across random seeds and reported with dispersion: every number here comes from one seed and one window, and the effects discussed are within the range seed variation alone can produce. Third, turnover must be constrained in the objective or charged in the evaluation, since the current protocol compares a daily-rebalancing agent with quarterly benchmarks at zero cost. Beyond those, extending the asset universe to include fixed-income instruments, currency futures, and real-estate investment trusts (FIIIs) would test whether the framework generalizes across asset classes. Second, incorporating realistic transaction costs and market-impact models would provide a more accurate picture of the net-of-cost performance improvement available to institutional investors. Third, replacing the rule-based SELIC regime classifier with a real-time macro nowcasting model or a hidden Markov Model could reduce the regime-labeling lag and further improve performance during transition periods. Finally, exploring alternative offline RL algorithms—such as Decision Transformer (Chen et al., 2021) or IQL (Kostrikov et al., 2021)—within the same framework would broaden the evidence base for offline RL as a tool for financial portfolio optimization in emerging markets.

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